MERN Mock Test

Decision Journal Application

Objective

Build a Decision Journal application where users can record important decisions they make and later evaluate the outcome of those decisions. The application should allow users to create, view, update, and delete decisions, and analyze how successful their decisions have been.

The goal is to assess your ability to implement CRUD operations, API design, database relationships, and simple analytics using the MERN stack.

Requirements

1. Decision Management (CRUD)

Users should be able to record decisions they make.

Each decision should contain the following fields:

Title

Description

Category (Career, Finance, Personal, Health, etc.)

Expected Outcome

Decision Date

Operations

Implement the following operations:

Create a new decision

View all decisions

View a single decision

Update a decision

Delete a decision

2. Decision Outcome (Evaluation)

After some time, the user should be able to record the actual outcome of a decision.

Each outcome should include:

Decision ID

Result (Success / Failure)

Rating (1–5)

Lessons Learned

Outcome Date

Operations

Add an outcome for a decision

Update an outcome

Delete an outcome

Each decision can have only one outcome record.

3. Insights / Analytics

Implement the following insights:

1. Decision Success Rate

Display the percentage of successful decisions.

Formula:

Success Rate = (Number of successful decisions / Total evaluated decisions) × 100

2. Most Successful Category

Display the category with the highest number of successful decisions.

Example output:

Most Successful Category: Finance

3. Average Decision Rating

Calculate the average rating of all evaluated decisions.

Example:

Average Rating: 3.8 / 5

Technical Requirements

Backend

Use:

Node.js

Express.js

MongoDB

Mongoose

Implement REST APIs such as:

POST /decisions

GET /decisions

GET /decisions/:id

PUT /decisions/:id

DELETE /decisions/:id

POST /outcomes

PUT /outcomes/:id

DELETE /outcomes/:id

GET /analytics

Database Design

Decisions Collection

```
{
  title: String,
  description: String,
  category: String,
  expectedOutcome: String,
  decisionDate: Date
}
```

Outcomes Collection

```
{
  decisionId: ObjectId,
  result: String,
  rating: Number,
  lessonsLearned: String,
  outcomeDate: Date
}
```

Frontend

Use React.js.

Create the following pages/components:

Decision List Page

Add Decision Form

Decision Details Page

Add Outcome Form

Analytics Dashboard

Expected Features

Display all decisions in a list

Ability to add and update decisions

Add outcome to a decision

View insights on dashboard

Basic validation